Chapter 11 solutions

CH11-E1

1. Without pivoting, the first pivot is ϵ, and

$$\ell_{21} = \frac{1}{\epsilon}$$

can be large. Partial pivoting compares $|\epsilon|$ and 1. For $0 < \epsilon \ll 1$, swap the rows, giving pivot 1 and

$$\ell_{21} = \epsilon.$$

2. Without pivoting, exact elimination gives

$$\epsilon x_1 + x_2 = 1,$$

Eliminating x_1 from the second equation yields

$$\left(1 - \frac{1}{\epsilon}\right) x_2 = 2 - \frac{1}{\epsilon}.$$

Setting $D = (1 - \epsilon)/\epsilon$ gives

$$x_2 = \frac{1 - 2\epsilon}{1 - \epsilon},$$
$$x_1 = \frac{1 - x_2}{\epsilon} = \frac{1}{1 - \epsilon}.$$

Pivoting only reorders equations and preserves the exact solution.

3. For large $1/\epsilon$, forming $1 - 1/\epsilon$, $2 - 1/\epsilon$, then subtracting can lose significant relative digits. Pivoting limits multiplier growth and reduces error amplification.

4. $\det A = \epsilon - 1 \approx -1$. This example is not becoming nearly dependent. As $\epsilon \to 0$, the matrix tends to $\begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$. Call it A_0. The matrix $A_0^T A_0 = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$ has eigenvalues $(3 \pm \sqrt{5})/2$, so

$$\kappa_2(A_0) = \sqrt{\frac{3 + \sqrt{5}}{3 - \sqrt{5}}} = \sqrt{\frac{7 + 3\sqrt{5}}{2}} \approx 2.618.$$

unpivoted elimination can be unstable on a well-conditioned problem. A determinant is not a general scale-invariant sensitivity measure, but that does not make this particular example ill-conditioned.

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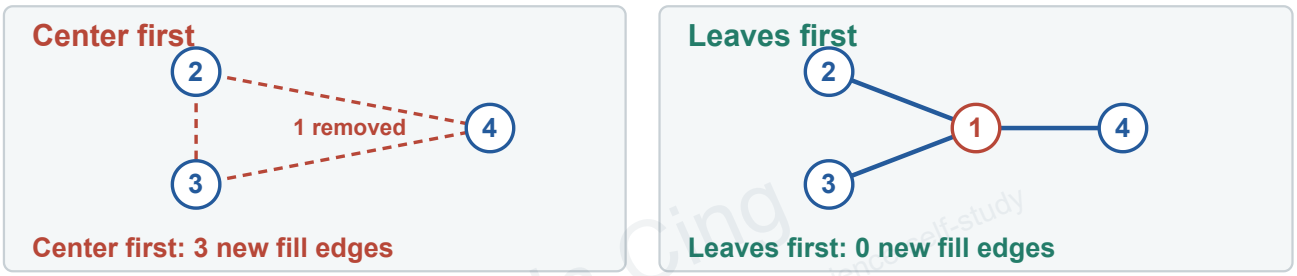
CH11-E2

1. Leading costs:

$$LU : \frac{2}{3}n^3, \qquad \text{forward solve} : n^2, \qquad \text{back solve} : n^2.$$

2. Refactoring separately repeats cubic work 1000 times. One factorization followed by quadratic solves avoids that repetition.
3. Eliminating a pivot couples its remaining neighbors, potentially creating cascading fill and nearly dense factors.
4. Center-first elimination makes a clique of leaves. Leaves-first elimination preserves the star's sparse structure. Parallel communication also affects ordering choices.

Elimination ordering: fill edges in a sparse star



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CH11-E3

1. Leading minors:

$$\Delta_1 = 4 > 0, \qquad \Delta_2 = 4(3) - 1 = 11 > 0,$$

Expand along row 1.

$$\Delta_3 = 4(15) - 1(5) + 2(-6) = 60 - 5 - 12 = 43 > 0.$$

All leading minors are positive, proving positive definiteness.

2. Let

$$R = \begin{bmatrix} r_{11} & 0 & 0 \\ r_{21} & r_{22} & 0 \\ r_{31} & r_{32} & r_{33} \end{bmatrix}.$$

Matching entries gives

$$r_{11} = 2, \quad r_{21} = 1/2, \quad r_{22} = \sqrt{3 - 1/4} = \frac{\sqrt{11}}{2}.$$

Off-diagonal:

$$r_{31} = 1, \qquad r_{32} = \frac{0 - r_{21} r_{31}}{r_{22}} = -\frac{1}{\sqrt{11}}.$$

$$r_{33} = \sqrt{5 - 1 - \frac{1}{11}} = \sqrt{\frac{43}{11}}.$$

Therefore

$$R = \begin{bmatrix} 2 & 0 & 0 \\ 1/2 & \sqrt{11}/2 & 0 \\ 1 & -1/\sqrt{11} & \sqrt{43/11} \end{bmatrix}.$$

3. Symmetry avoids separately computing both triangles. The leading count is $n^3/3$ rather than $2n^3/3$, under the same dense arithmetic model.

4. Multiplication by 10^{-8} scales all singular values by that factor, so

$$\kappa_2(A') = \kappa_2(A).$$

Scaling A, b together leaves x and relative conditioning unchanged. Scaling only A multiplies the solution by 10^8. Absolute accuracy and physical units remain separate issues.

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CH11-E4

1. For $y = Ax$,

$$\|y\|_1 \le \sum_j \left(\sum_i |a_{ij}| \right) |x_j| \le \max_j c_j \|x\|_1.$$

Choose $x = e_j$ at a column attaining the largest absolute column sum. Equality is attained, proving the formula.

2.

$$|(Ax)_i| \le \left(\sum_j |a_{ij}| \right) \|x\|_\infty.$$

Let i_* index a maximum absolute row sum. For a real matrix, choose $x_j = \operatorname{sgn}(a_{i_* j})$, taking $x_j = 1$ at zero entries. Then $\|x\|_\infty = 1$ and $(Ax)_{i_*} = \sum_j |a_{i_* j}|$, attaining the upper bound. For a complex matrix, choose $x_j = \overline{a_{i_* j}}/|a_{i_* j}|$ at nonzero entries, again taking 1 at zeros. The same attainment proves $\|A\|_\infty = \max_i r_i$.

3. For

$$A = \begin{bmatrix} 1 & 2 \\ -3 & 4 \end{bmatrix},$$

$$\|A\|_1 = \max(4, 6) = 6,$$

$$\|A\|_\infty = \max(3, 7) = 7,$$

$$\|A\|_F = \sqrt{1 + 4 + 9 + 16} = \sqrt{30}.$$

4. For an invertible square matrix and an induced norm,

$$1 = \|I\| = \|AA^{-1}\| \le \|A\|\|A^{-1}\| = \kappa(A).$$

This includes the Euclidean case.

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CH11-E5

1. $A = 10^{-8} I$, $b = (1, 1)^T$, $\widehat{x} = 0$:

$$r = b, \qquad \|r\|_2 = \sqrt{2}.$$

The exact solution is $(10^8, 10^8)^T$, giving

$$\|e\|_2 = \sqrt{2} \cdot 10^8.$$

2. Relative error:

$$\frac{\|e\|_2}{\|x\|_2} = 1.$$

Normalized backward error:

$$\eta = \frac{\sqrt{2}}{10^{-8} \cdot 0 + \sqrt{2}} = 1.$$

3. This matrix has condition number 1 and is not ill-conditioned. Its residual $\sqrt{2}$ is not small, while the exact solution is large because the matrix scale is small. Scaling both sides can arbitrarily shrink the absolute residual without changing the solution or relative error. Use scale-aware residuals or backward error.

4. Right-hand side perturbation:

$$\frac{\|\Delta x\|_2}{\|x\|_2} \leq \kappa_2(A) \frac{\|\Delta b\|_2}{\|b\|_2}.$$

Here $\kappa_2(A) = 1$. Scaling A, b by 10^8 gives an identity coefficient matrix with the same solution, clarifying why absolute tolerances cannot be compared across unreported scales.

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CH11-E6

1. $A = D + L + U$,

$$G_J = -D^{-1}(L+U) = \begin{bmatrix} 0 & -1/4 \\ -1/3 & 0 \end{bmatrix}.$$

Component updates:

$$x_1^{(k+1)} = \frac{1 - x_2^{(k)}}{4}, \qquad x_2^{(k+1)} = \frac{2 - x_1^{(k)}}{3}.$$

2.

$$G_J^2 = \begin{bmatrix} 1/12 & 0 \\ 0 & 1/12 \end{bmatrix}.$$

The eigenvalues of the iteration matrix are $\pm 1/\sqrt{12}$, so

$$\rho(G_J) = \frac{1}{\sqrt{12}} < 1,$$

the iteration converges.

3. $x_0 = 0$:

$$x_1 = \left(\frac{1}{4}, \frac{2}{3}\right)^T.$$

The second iterate is

$$x_2 = \left(\frac{1 - 2/3}{4}, \frac{2 - 1/4}{3}\right)^T = \left(\frac{1}{12}, \frac{7}{12}\right)^T.$$

4. Spectral radius determines asymptotic convergence, not monotonic decay of every component. A norm with $\|G_J\| < 1$ does give contraction. Here $\|G_J\|_\infty = 1/3$, so that error norm contracts, although one zero error component can become nonzero on the next step. SPD alone does not guarantee general Jacobi convergence.

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CH11-E7

1.

$$\phi(x) = \frac{1}{2}x^T Ax - b^T x = x_1^2 + x_1 x_2 + x_2^2 - 3x_1 - 3x_2.$$

$$\nabla \phi = Ax - b.$$

2. $x_0 = 0$, $r_0 = b = (3,3)^T$, $p_0 = r_0$.Line-search step:

$$\alpha_0 = \frac{r_0^T r_0}{p_0^T A p_0}.$$

$$p_0^T A p_0 = (3,3)\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}\begin{bmatrix} 3 \\ 3 \end{bmatrix} = 54.$$

Since $r_0^T r_0 = 18$, the step length is $1/3$.

3.

$$x_1 = \alpha_0 p_0 = (1,1)^T.$$

Now $Ax_1 = b$, completing the exact solve in one step.

4. Finite termination relies on exact inner products and conjugacy. Rounding destroys these identities, and an ill-conditioned floating-point problem can need more than n iterations. A stopping tolerance may also accept a result before the forward error is small.

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CH11-E8

1. Let

$$AM^{-1}y = b, \qquad x = M^{-1}y.$$

Then $Ax = b$. Conversely, an original solution gives $y = Mx$ when M is invertible.

2. When A is invertible, choosing $M = A$ makes the transformed matrix the identity: in exact arithmetic GMRES takes one step for a nonzero initial residual and stops immediately for a zero initial residual. However, applying its inverse requires solving the original problem, so it is not an inexpensive preconditioner. Factor reuse may still be useful for other reasons, but it does not remove the original solve cost.

3. For singular M, M^{-1} does not exist and $x = M^{-1}y$ cannot establish a bijection. Substituting a pseudoinverse introduces range and nullspace restrictions and requires a separate equivalence proof. For invertible but nearly singular M, errors in applying M^{-1} may be amplified, and an unstable preconditioner solve can impair iteration. Symbolic invertibility is not numerical stability.

4. Nonnormal matrices can have substantial transients despite eigenvalues in the unit disk. GMRES depends on residual polynomials, Krylov spaces, and nonnormal geometry, not just eigenvalue magnitudes.

5. Initially $\beta = \sqrt{2}$, $v_1 = (1,1)^T/\sqrt{2}$. The first product is $w = Av_1 = (2,1)^T/\sqrt{2}$, giving $h_{11} = 3/2$. Subtraction leaves $(1,-1)^T/(2\sqrt{2})$, hence $h_{21} = 1/2$, $v_2 = (1,-1)^T/\sqrt{2}$, and $\overline{H}_1 = (3/2, 1/2)^T$. Solve

$$\min_y (\sqrt{2} - 3y/2)^2 + (y/2)^2.$$

Differentiation gives $5y - 3\sqrt{2} = 0$, so $y = 3\sqrt{2}/5$ and $x_1 = (3/5, 3/5)^T$. Check $Ax_1 = (6/5, 3/5)^T$, $r_1 = (-1/5, 2/5)^T$, and $\|r_1\|_2^2 = 1/5$, with $(Av_1)^T r_1 = 0$.

The second product is $Av_2 = (0,-1)^T/\sqrt{2}$. First $h_{12} = -1/2$; subtraction leaves $(1,-1)^T/(2\sqrt{2})$. Next $h_{22} = 1/2$, and the remaining vector is zero. Thus $h_{32} = 0$, with

$$H_2 = \begin{bmatrix} 3/2 & -1/2 \\ 1/2 & 1/2 \end{bmatrix}, \qquad H_2 y = \begin{bmatrix} \sqrt{2} \\ 0 \end{bmatrix}.$$

Row two gives $y_2 = -y_1$; row one gives $2y_1 = \sqrt{2}$. Therefore $y = (1,-1)^T/\sqrt{2}$ and $x_2 = V_2 y = (0,1)^T$. Directly, $Ax_2 = (1,1)^T = b$. The zero true residual establishes a happy breakdown. Do not form the undefined $v_3 = w/h_{32}$.

CH11-E9

Record the following quantities.

1. Shapes, data types, and norms of A, b.
2. Factorization or preconditioner type and ordering.
3. Initial and final true residuals.
4. Relative residual and backward-error estimate.
5. condition estimate;
6. Iteration count, convergence flag, and time limit.
7. Results for a second ordering or initial guess.
8. Deterministic settings or random seed when relevant.

The final true residual checks whether the recurrence still represents the original equation. A condition estimate explains the possible forward error at that residual. Agreement between two solvers is useful evidence but does not replace stated tolerances and assumptions.

CH11-E10

1. Symmetry is immediate. The leading minors are

$$\Delta_1 = 4 > 0, \qquad \Delta_2 = 4 \cdot 3 - 1 = 11 > 0,$$

so A is positive definite. Writing $A = RR^T$ gives

$$R = \begin{bmatrix} 2 & 0 \\ 1/2 & \sqrt{11}/2 \end{bmatrix}.$$

Forward and back substitution give

$$x = R^{-T}R^{-1}b = (1/11, 7/11)^T,$$

and substitution yields $Ax = (1, 2)^T$.

CH11-E10 (continued)

2. The eigenvalues are

$$\lambda_{\pm} = \frac{7 \pm \sqrt{5}}{2}.$$

For SPD matrices, the 2-norm is the largest eigenvalue and the inverse norm is the reciprocal of the smallest, so

$$\|A\|_2 = \frac{7+\sqrt{5}}{2}, \quad \|A^{-1}\|_2 = \frac{2}{7-\sqrt{5}}, \quad \kappa_2(A) = \frac{7+\sqrt{5}}{7-\sqrt{5}} \approx 1.93875.$$

Starting from $x_0 = 0$, CG satisfies

$$\|x - x_k\|_A \le 2q^k \|x\|_A, \qquad q = \frac{\sqrt{\kappa_2(A)} - 1}{\sqrt{\kappa_2(A)} + 1} \approx 0.16402.$$

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CH11-E10 (continued)

3.

$$A\widehat{x} = \begin{bmatrix} 1 \\ 1.9 \end{bmatrix}, \qquad r = b - A\widehat{x} = (0, 0.1)^T.$$

$$e = x - \widehat{x} = (-1/110, 2/55)^T \approx (-0.0090909, 0.0363636)^T.$$

Hence

$$\|r\|_2 = 0.1, \qquad \|e\|_2 \approx 0.03748, \qquad \frac{\|e\|_2}{\|x\|_2} \approx 0.05831.$$

This also agrees with $\|e\|_2 \le \|A^{-1}\|_2 \|r\|_2 \approx 0.041982$. The normwise backward error uses

$$\eta = \frac{\|r\|_2}{\|A\|_2 \|\widehat{x}\|_2 + \|b\|_2}$$

giving $\eta \approx 0.01982$. Residual and forward error are different quantities linked by conditioning.

4. $H^H = H$, and

$$\begin{vmatrix} 2 & 1+i \\ 1-i & 3 \end{vmatrix} = 6 - |1+i|^2 = 4 > 0,$$

together with first leading minor $2 > 0$, proves Hermitian positive definiteness. The characteristic polynomial is

$$\det(H - \lambda I) = \lambda^2 - 5\lambda + 4 = (\lambda - 1)(\lambda - 4),$$

so the eigenvalues are $1, 4$, and $\kappa_2(H) = 4/1 = 4$.

$$b_c = Hx_c = (4 + 2i, 7 - i)^T.$$

With $x_0 = x_c$,

$$r_0 = b_c - Hx_0 = 0.$$

first check $\|r_0\|_2 \le \tau$. Here it vanishes, so return x_0 after zero iterations rather than computing $(0^H 0)/(p_0^H A p_0)$.

5. The complex inner product is $u^H v$. Without conjugation, $r^T r$ may be complex and is not a nonnegative squared norm; the purported error energy may also become complex. Since $H^T = \overline{H} \neq H$, ordinary transpose is not the Hermitian test. Confusing a general complex symmetric matrix with a Hermitian one loses the positive-real-energy guarantee. However, this particular H^T is still Hermitian positive definite with eigenvalues 1 and 4. Replacing H by H^T solves a different system: $H^T x_c = (4 - 2i, 7 + i)^T \neq b_c$. Its convergence cannot certify the original equations.

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CH11-E11

1. The leading minors are 2, 3, 4, proving positive definiteness. Start with $r_0 = p_0 = (1,0,0)^T$. The steps are

k	p_k	Ap_k	α_k	x_{k+1}	r_{k+1}	β_k
0	$(1,0,0)$	$(2,1,0)$	$1/2$	$(1/2,0,0)$	$(0,-1/2,0)$	$1/4$
1	$(1/4,-1/2,0)$	$(0,-3/4,-1/2)$	$2/3$	$(2/3,-1/3,0)$	$(0,0,1/3)$	$4/9$
2	$(1/9,-2/9,1/3)$	$(0,0,4/9)$	$3/4$	$(3/4,-1/2,1/4)$	0	terminated

   Denominators $p_k^T Ap_k$ are $2, 3/8, 4/27$; numerators $\|r_k\|^2$ are $1, 1/4, 1/9$, giving the listed step lengths. Check $A(3/4,-1/2,1/4)^T = (1,0,0)^T$.

2. The three nonzero residuals lie on distinct standard axes and have zero pairwise ordinary inner products. Direction conjugacy is

$$p_0^T Ap_1 = 0, \quad p_0^T Ap_2 = 0, \quad p_1^T Ap_2 = (1/4,-1/2,0)\cdot(0,0,4/9) = 0.$$

but $p_0^T p_1 = 1/4 \neq 0$. CG requires orthogonality in the energy inner product, not ordinary orthogonality of directions.

3. For $z \in \mathcal{K}_k$, $\phi(x_k+z) - \phi(x_k) = -z^T r_k + z^T Az/2 = z^T Az/2$, strictly positive for nonzero z, proving a unique minimum. For $i \leq k-2$, $Ap_i \in \mathcal{K}_k$, so residual orthogonality already gives $p_i^T Ar_k = 0$. Choose β only to make $p_k = r_k + \beta_{k-1} p_{k-1}$ conjugate to the previous direction. The residual update and adjacent residual orthogonality give

$$Ap_{k-1} = \frac{r_{k-1} - r_k}{\alpha_{k-1}}, \quad p_{k-1}^T Ar_k = -\frac{\|r_k\|^2}{\alpha_{k-1}}, \quad p_{k-1}^T Ap_{k-1} = \frac{\|r_{k-1}\|^2}{\alpha_{k-1}}.$$

Hence

$$\beta_{k-1} = -\frac{p_{k-1}^T Ar_k}{p_{k-1}^T Ap_{k-1}} = \frac{\|r_k\|^2}{\|r_{k-1}\|^2}.$$

Line minimization makes the new residual orthogonal to the new direction. For earlier directions, $r_{k+1}^T p_i = r_k^T p_i - \alpha_k p_k^T Ap_i = 0$. A nonzero r_k, orthogonal to $\mathcal{K}_k$, extends the search space. Nonzero conjugate directions are independent, so there are at most n. Once they span the whole space, residual orthogonality forces zero residual.

CH11-E12

1. $\|A\|_2 = 1$, $\|A^{-1}\|_2 = 1/\varepsilon$, and $\kappa_2(A) = 1/\varepsilon$. The inverse difference is

$$(A+E)^{-1} - A^{-1} = \operatorname{diag}\left(0, \frac{1}{\varepsilon-\delta} - \frac{1}{\varepsilon}\right) = \operatorname{diag}\left(0, \frac{\delta}{\varepsilon(\varepsilon-\delta)}\right).$$

The relative change is $\delta/(\varepsilon-\delta)$. The relative input perturbation is δ, and $\kappa\delta/(1-\kappa\delta)$ equals this value. Reducing the smallest singular value attains the bound. At $\delta = \varepsilon$, the matrix is singular and the bound's invertibility premise fails.

2. Direct inner products give

$$Q^T A = \begin{bmatrix} \sqrt{2} & 1/\sqrt{2} \\ 0 & \sqrt{3/2} \\ 0 & 0 \end{bmatrix}, \qquad Q^T b = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{6} \\ -2/\sqrt{3} \end{bmatrix}.$$

Solve the top 2×2 block. The second equation gives $x_2 = 1/3$, and the first $x_1 = 1/3$. The prediction is $(1/3, 1/3, 2/3)^T$, residual $(2/3, 2/3, -2/3)^T$, and squared norm $4/3$, equal to the unfit component $|-2/\sqrt{3}|^2$. Full Q has three columns, so $Q^T b \in \mathbb{R}^3$ and only its first two entries form c_1. Reduced $Q_1 = [q_1, q_2]$ has two columns, so $Rx = Q_1^T b$ directly.

CH11-E13

$r_0=p_0=b$, $r_0^\top r_0=101/100$ and $p_0^\top A p_0=2$. Hence $\alpha_0=101/200$, $x_1=(101/200,101/2000)^\top$ and $r_1=(99/200,-99/20)^\top$. Its squared norm is $9801/40000+9801/400=989901/40000$, larger than $101/100$.

The exact solution is $x^*=(1,1/1000)^\top$. The initial energy gap is $\tfrac{1}{2}b^\top A^{-1}b=10001/20000$. Along p_0, exact line search decreases F by $(r_0^\top r_0)^2/(2p_0^\top A p_0)=10201/40000$. The remaining gap is $20002/40000-10201/40000=9801/40000$, strictly smaller than before. Energy minimization controls the A-norm of solution error, not monotone decrease of the Euclidean residual norm.

For the second matrix, $Ap_1=(-11/8,11/16)^\top$, so $p_0^\top A p_1=-11/8+22/16=0$. Ordinary inner product gives $-7/16+6/8=5/16\neq 0$. Residuals $r_0=(1,2)^\top$ and $r_1=(-1/2,1/4)^\top$ instead satisfy $-1/2+2/4=0$. These are different pairs of vectors and different inner products.

References and attribution

Lin Cing, handwritten linear algebra notes. Gilbert Strang, Introduction to Linear Algebra, 5th ed., Chapter 11.

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